Experiment 5

Aim: To study the structure of a Solidity program, understand its basic constructs, and perform compilation and deployment of a smart contract using the Remix IDE environment.

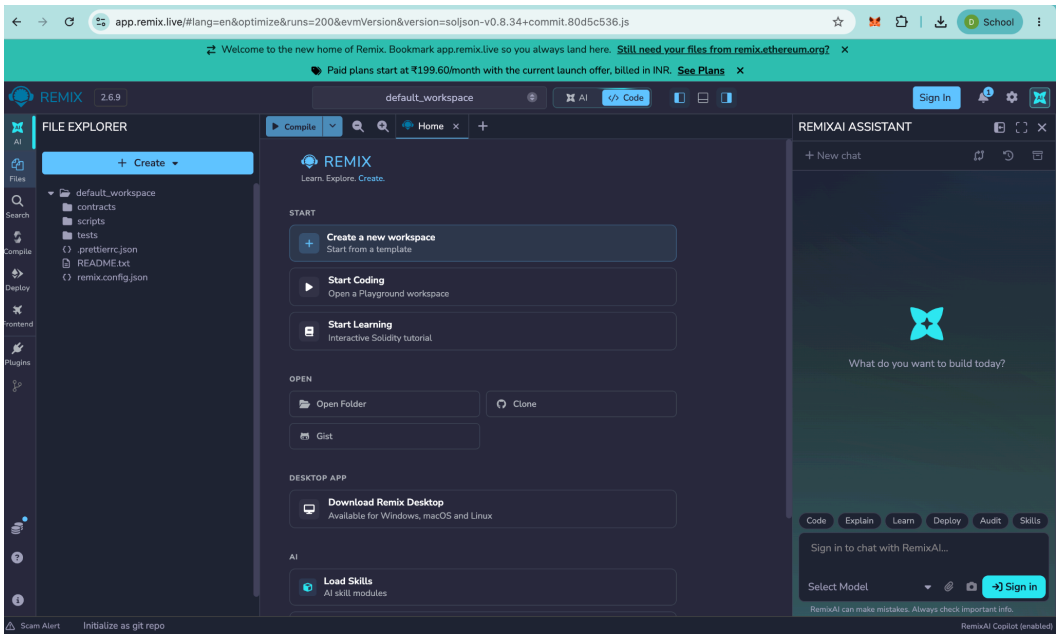
Theory: Solidity is a high-level, contract-oriented programming language used to develop smart contracts on Ethereum-compatible blockchains.

A smart contract is a program stored and executed on a blockchain. It can store data and perform operations according to predefined rules.

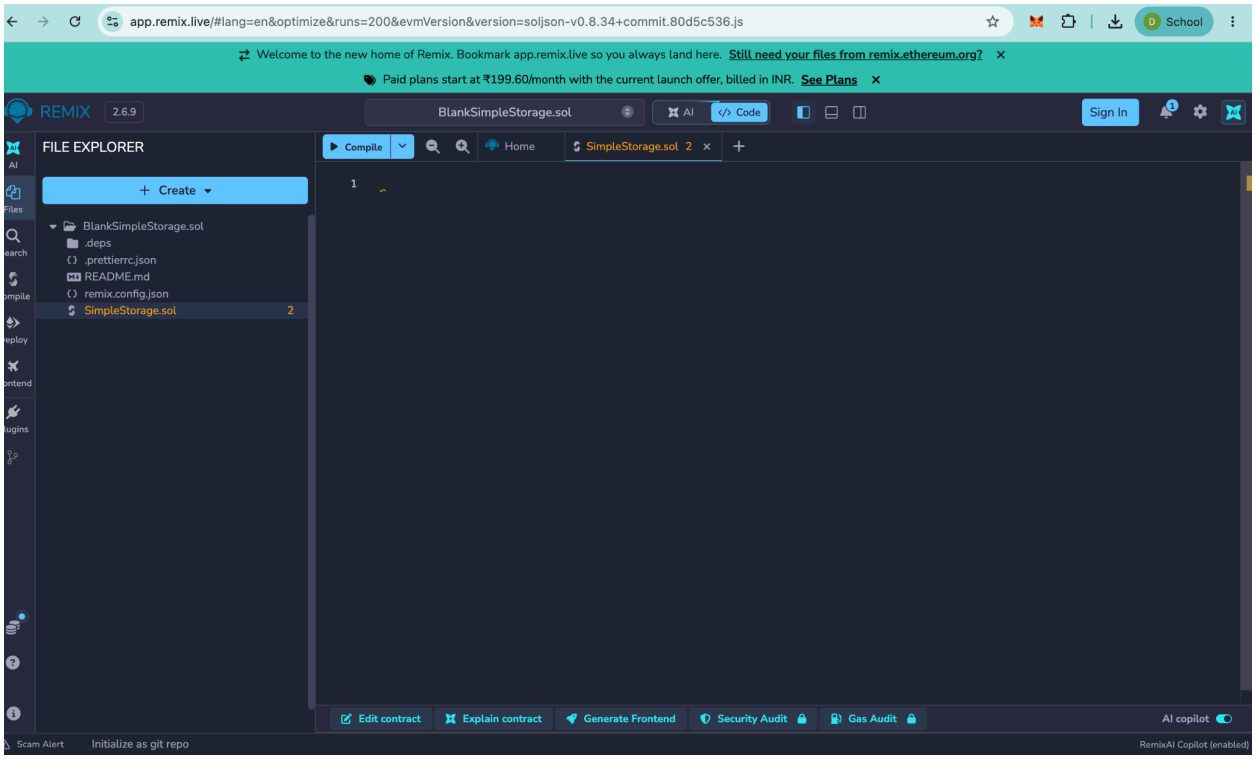
The main stages of smart contract development are:

1. Write the Solidity program.
2. Compile the program into bytecode.
3. Deploy the compiled contract to a blockchain environment.
4. Interact with the deployed contract using its functions.

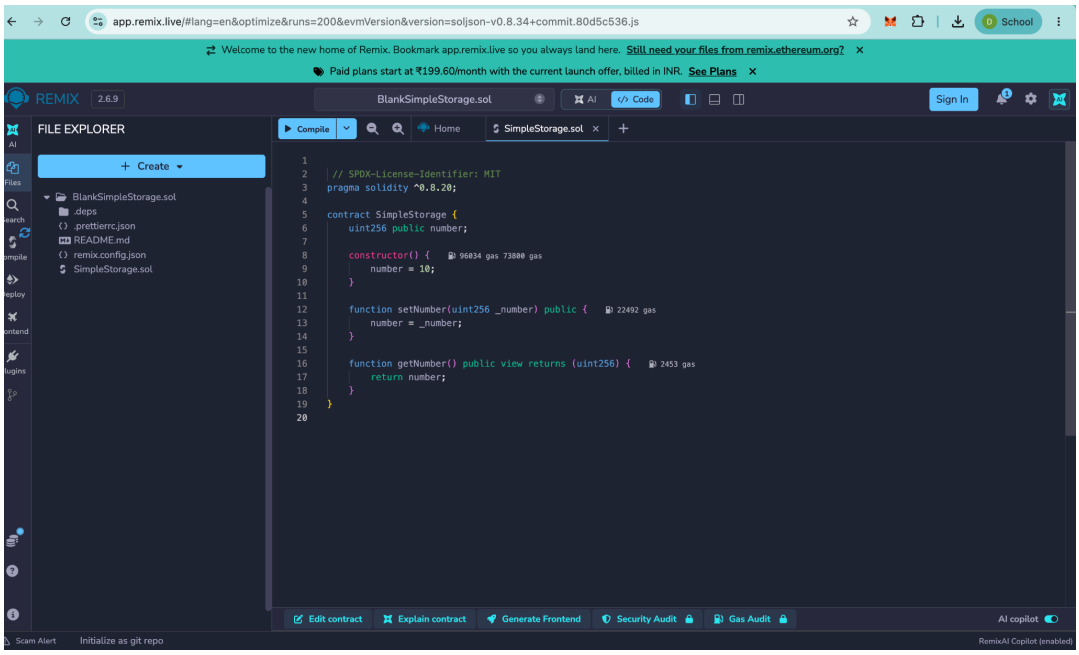
Step 1: Open Remix IDE



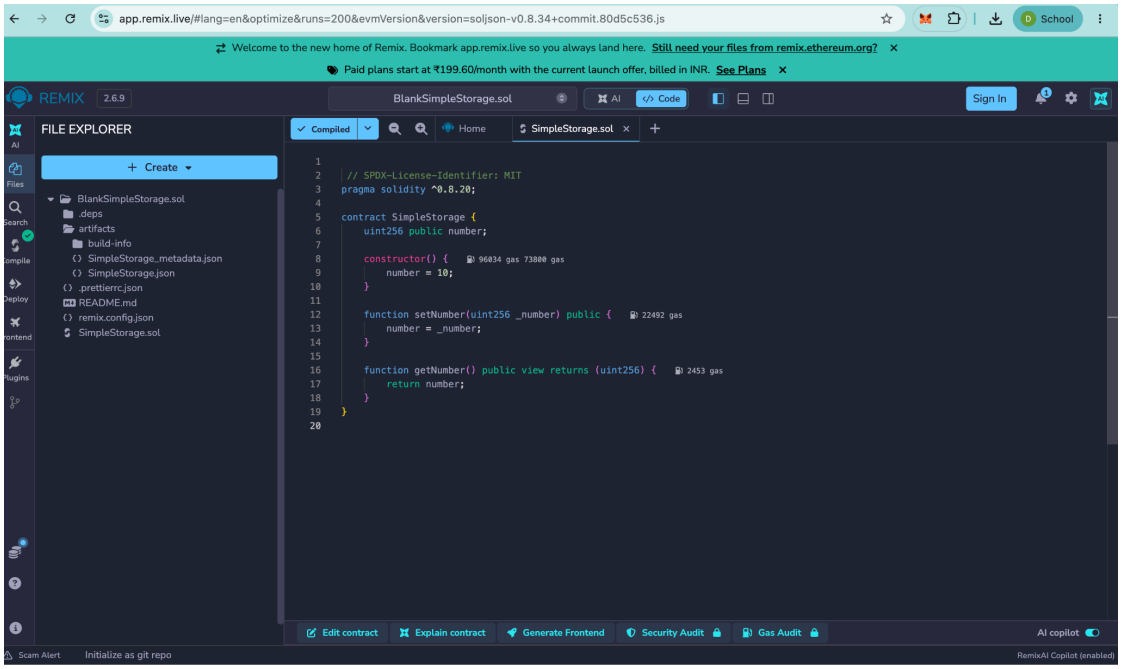
Step 2: Create the Solidity file



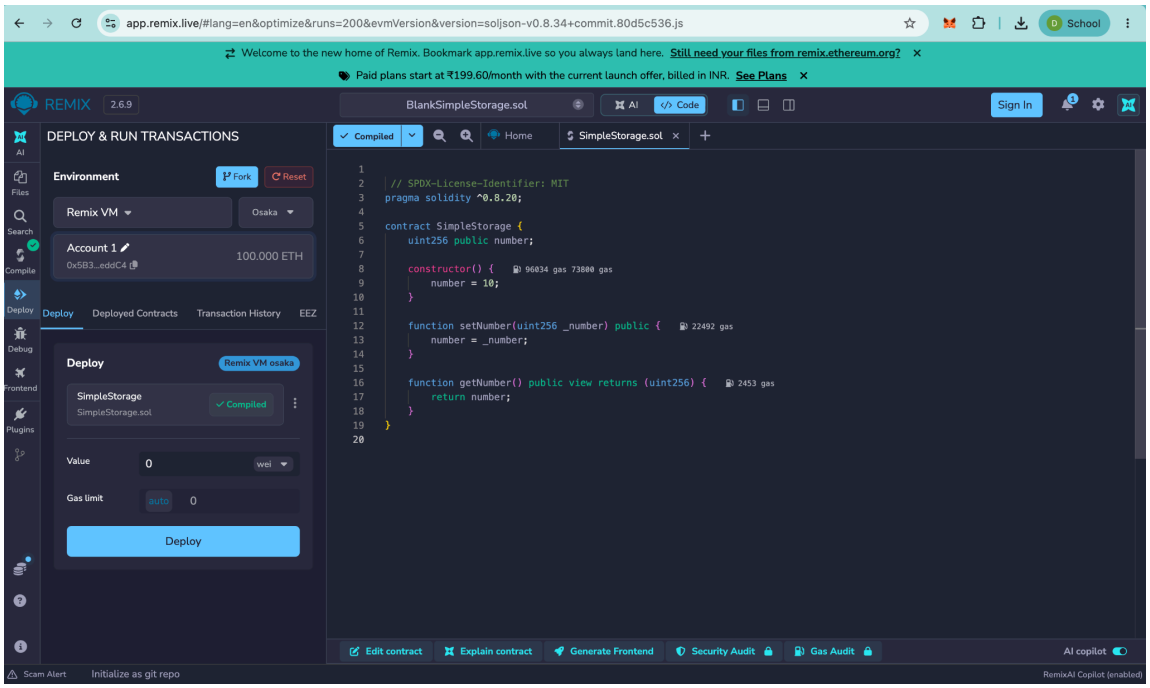
Step 3: then Solidity code



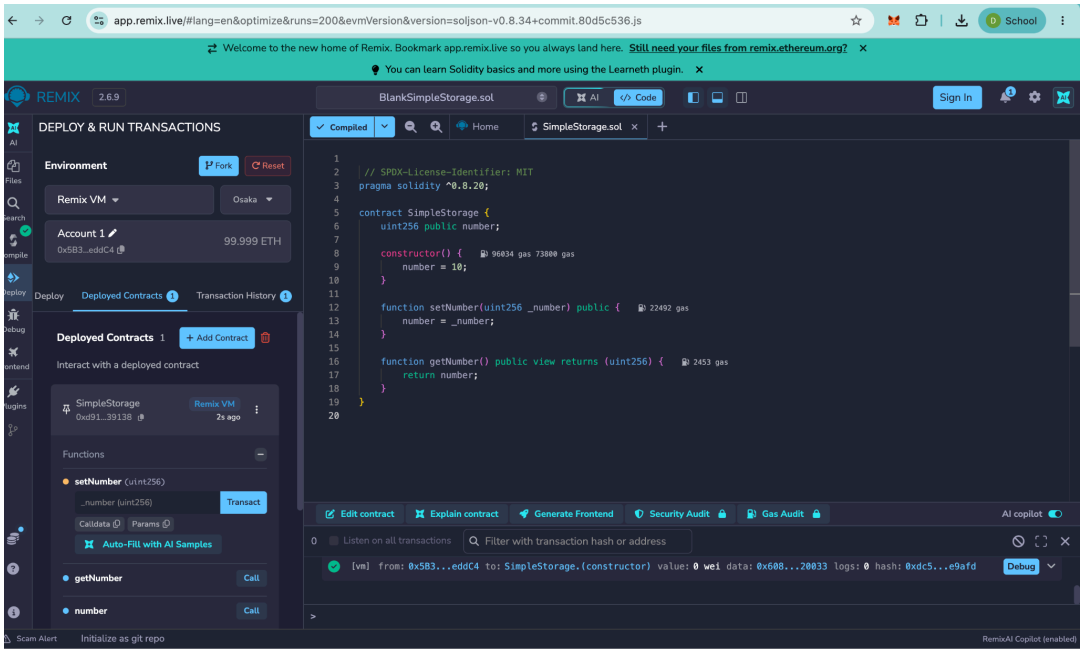
Step 4: Compile the code



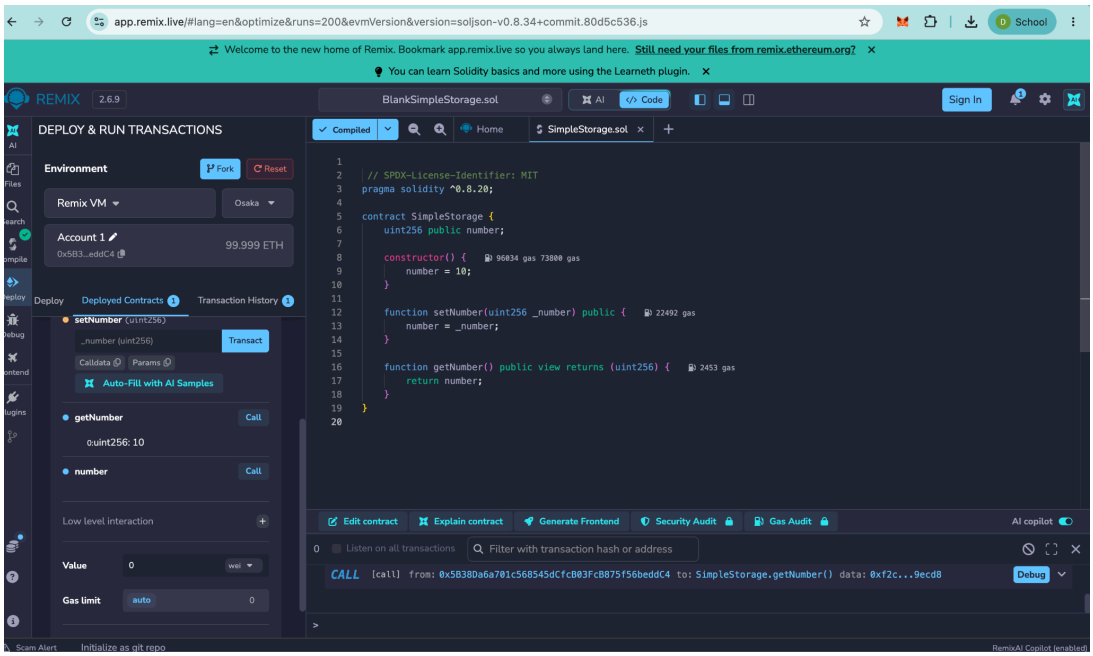
Step 5: Deploy the smart contract



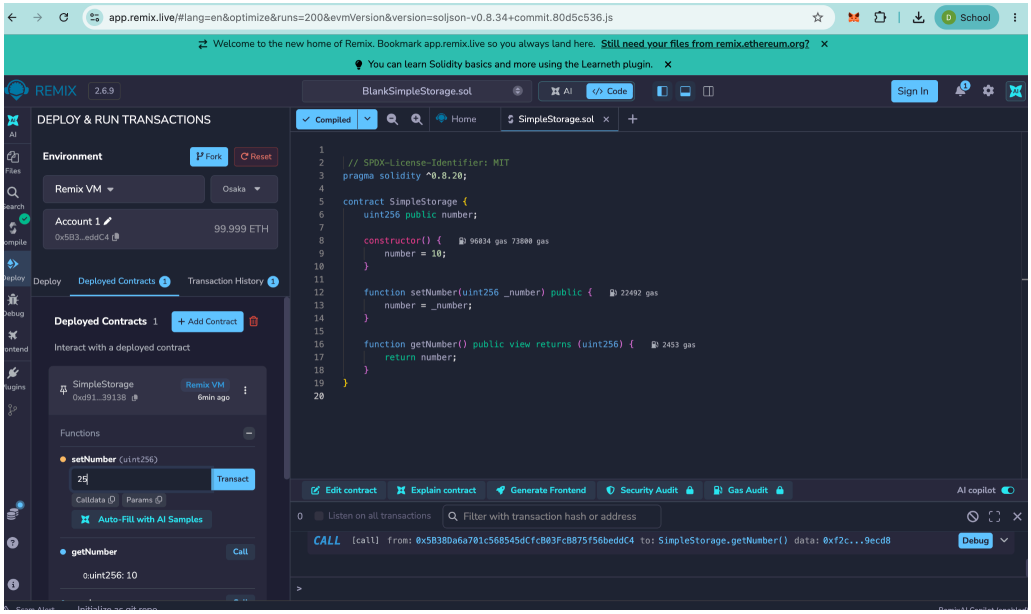
Step 6: Check deployment

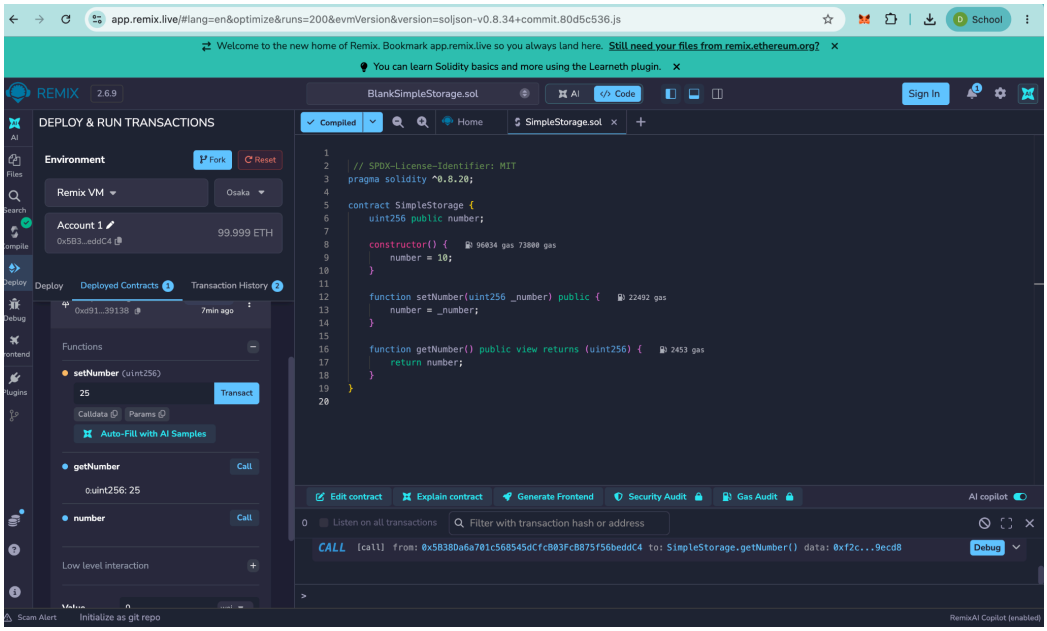


Step 7: Test the contract



Step 8: Test the contract by changing the value from 10 to 25





observation:The Solidity smart contract was compiled and deployed successfully using Remix IDE. The initial value of the variable `number` was 10. After executing the `setNumber()` function with the value 25, the stored value was updated to 25 and verified using `getNumber()`.

Result:The structure and basic constructs of Solidity were studied. Compilation, deployment, and execution of a smart contract were successfully performed using Remix IDE.

Conclusion:Solidity provides the basic constructs required to develop smart contracts. Remix IDE provides a convenient environment for writing, compiling, deploying, and testing smart contracts using a simulated blockchain.